

Operating System CoSc2032

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1. Introduction

CoSc2032 Operating System

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Introduction

- What is an operation system?
- History of Operating Systems
- OS Services and OS Structure

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What is an Operating System?

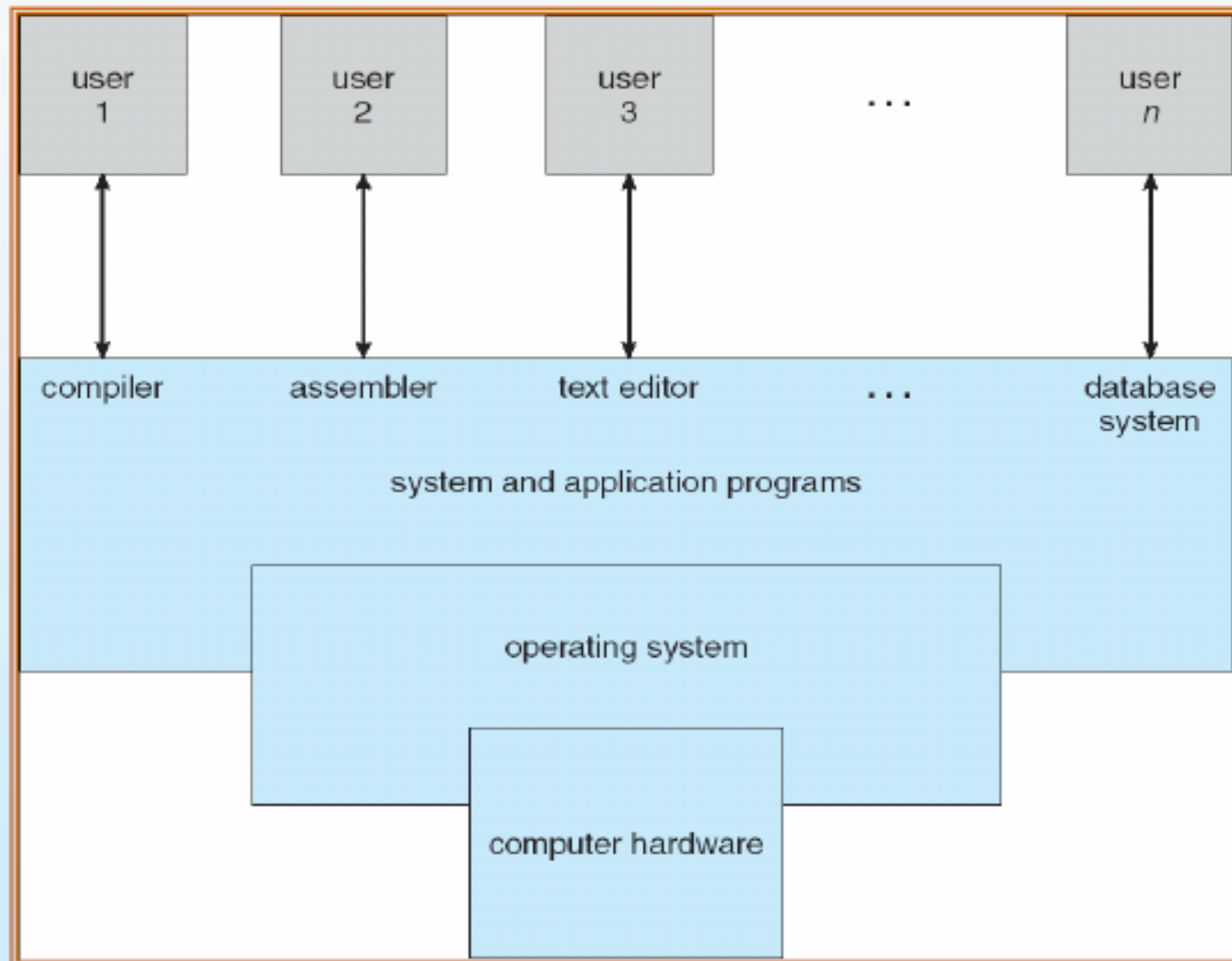
- An operating system is a program that manages the computer hardware.
- It also provides a basis for application programs and acts as an intermediary between the computer user and computer hardware.

What do OS Do

The computer system can be divided roughly into four components:

- The hardware
- The operating system
- The application software
- The Users

Four Components of a Computer System



History of OS

- The First Generation (1945 – 55)
 - Vacuum Tubes and Plugboards
 - Each machine designed, built, programmed, operated, and maintained a single group of people
 - Mode of operation:
 - Programmers sign up for a block of time
 - Programmers insert a plugboard into the computer in the machine room
 - Plugboard was replaced by punched card in early 1950s

History of OS

- The Second Generation (1955 – 65)
 - Transistors and Batch Systems
 - Computers became reliable enough to be used by customers (users)
 - Clear separation between professions
 - Designers, builders, operators, programmers and maintenance personnel

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History of OS

- The Second Generation (1955 – 65)
 - Mainframe
 - Job
 - Programming (FORTRAN)
 - Batch System

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History of OS

- The Second Generation (1955 – 65)



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History of OS

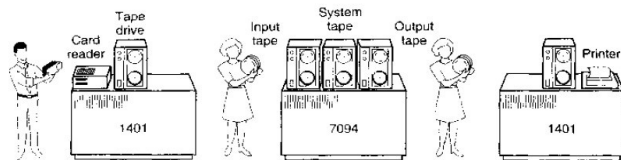
- The Second Generation (1955 – 65)



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History of OS

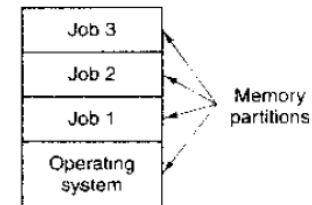
- The Second Generation (1955 – 65)



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History of OS

- The Third Generation (1965 – 80)
 - IC and Multi Programming



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History of OS

- The Third Generation (1965 – 80)
 - Spooling: ability of reading jobs from cards onto the disk
 - IBM 1401s were not needed
 - Much of carrying of tapes disappeared
 - However, 3rd generation systems were still batch system
 - Problems of batch system paved the way to timesharing

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History of OS

- The Third Generation (1965 – 80)
 - Timesharing

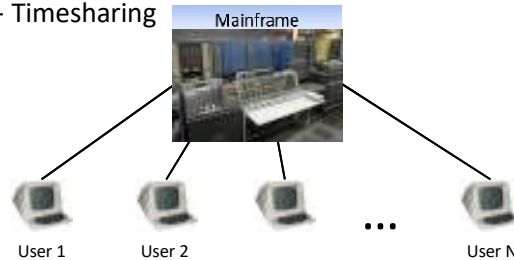


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History of OS

• The Third Generation (1965 – 80)

- Timesharing



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History of OS

• The Fourth Generation (1980 – ???)

- Personal Computers
- No basic difference between Micro and Mini
- Business drama
 - Intel's 8080 € Digital Research's CP/M (1977)
 - The deal between IBM PC and Bill Gate's BASIC Interpreter (1980s)
 - Seattle Computer Products (DOS)
 - DOS/BASIC
 - Time Paterson MS-DOS retrospect
 - Steve Jobs followed Doug Engelbart's GUI
 - Lisa € Macintosh
 - MS Windows influenced by Macintosh

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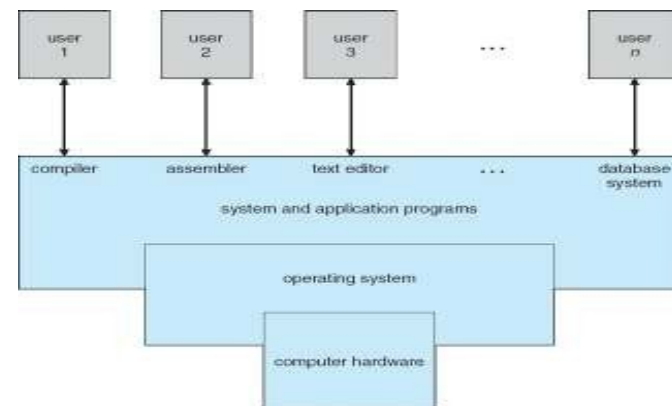
What is an operation system?

- A layer in the modern computer system that is responsible to manage several devices and provide user programs with a simpler interface to the hardware.

- Extended Machine
- Resource Manager

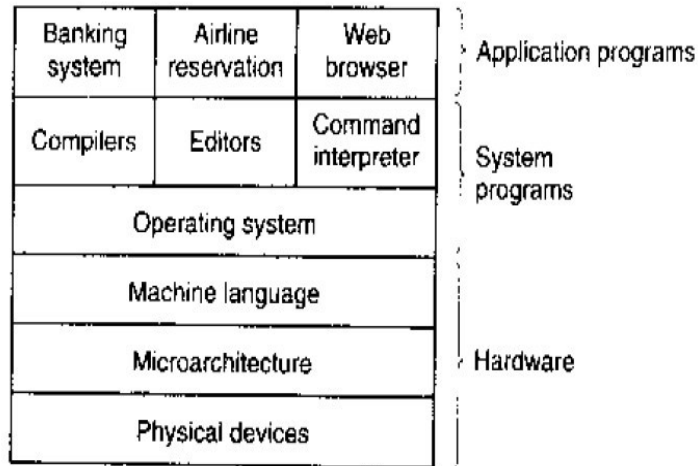
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What is an operation system?



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OS Structure and Services



OS Structure and Services

- **Kernel Mode**
 - Protected from user application tampering the hardware
- **User Mode**
 - The mode in which applications and system programs run
 - Permits only a subset of the instructions to be executed and a subset of features to be accessed
 - Disallow I/O instruction, Memory Protection, PSW Setting

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Reading Assignment

- **Architectures of Operating Systems**
 - Monolithic
 - Layered
 - Kernel
 - Microkernel
 - Virtual Machines
- **Increasing Efficiency**
 - Multi program
 - Multi User

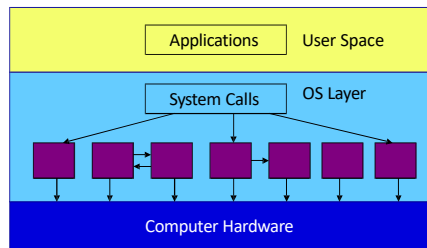
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Monolithic Architecture

- **Monolithic Architecture—the early operating systems**
 - Every component is contained in the kernel, can directly communicate with other components

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Monolithic Architecture



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Monolithic Architecture

- Pros
 - Highly efficient – by direct intercommunication between components
- Cons
 - difficult to develop
 - difficult to isolate the source of bugs and other errors
 - particularly susceptible to damage from malicious code

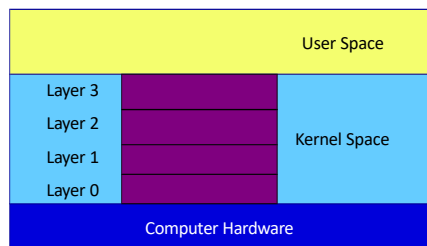
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Layered Architecture

- Layered OS structure:
 - Group components that perform similar functions into layers. Each layer communicates only with neighbour layer



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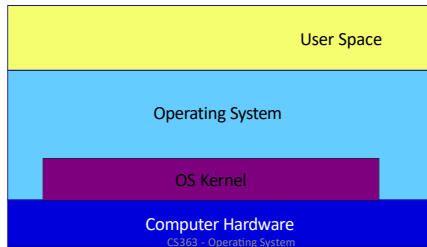
Layered Architecture

- Pros
 - It provides good modularity – helps simplify the development of an OS
- Cons
 - Less efficient
 - Complex design – each functionality has to be divided into parts to fit into different layers.

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Kernel Based Architecture

- It separates the machine-independent parts from the machine-dependent parts
 - Kernel is machine-dependent. It contains the basic component of OS.



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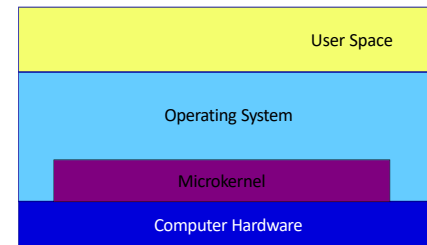
Kernel Based Architecture

- Pros
 - Better portability—Kernel encloses all the machine-dependent code
- Cons
 - Ensuring security and extensibility is a challenge

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Microkernel Based Architecture

- As OS expanded, the kernel became large and difficult to manage
 - Microkernel approach removes all nonessential components from the kernel and implementing them as system and user-level programs.
 - Result: A smaller kernel



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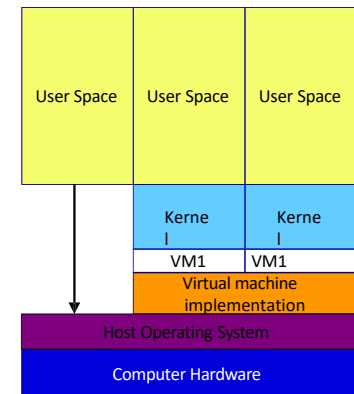
Microkernel Based Architecture

- Pros
 - Enhance portability, extensibility, reliability and security
- Cons
 - Less efficient—increased system function overhead

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Virtual Machines

- Can create the illusion that there are more than one separate machines.



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OS Structure and Services

- To obtain services from the operating system, a user program must make a system call
- The interface between the operating system and the user programs is defined by the set of system calls **that the operating system provides.**

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OS Structure and Services

- OS Services
 - Process Management
 - Memory Management
 - File Management
 - I/O Management
 - User Management

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